

— SILICATE LABS

PPay

Payment infrastructure for Pakistan. One API and one dashboard for cards, wallets, and bank transfers, with fraud scoring built into the payment path.

Getting paid in Pakistan takes four integrations and still leaves gaps

Fragmented rails

Cards go through one gateway, JazzCash and Easypaisa through another, bank transfer through a third. Each has its own API, its own dashboard, its own reconciliation.

Months, not hours

Legacy bank gateways are quoted in months of integration work. International platforms integrate faster but do not support the wallets most Pakistani customers actually use.

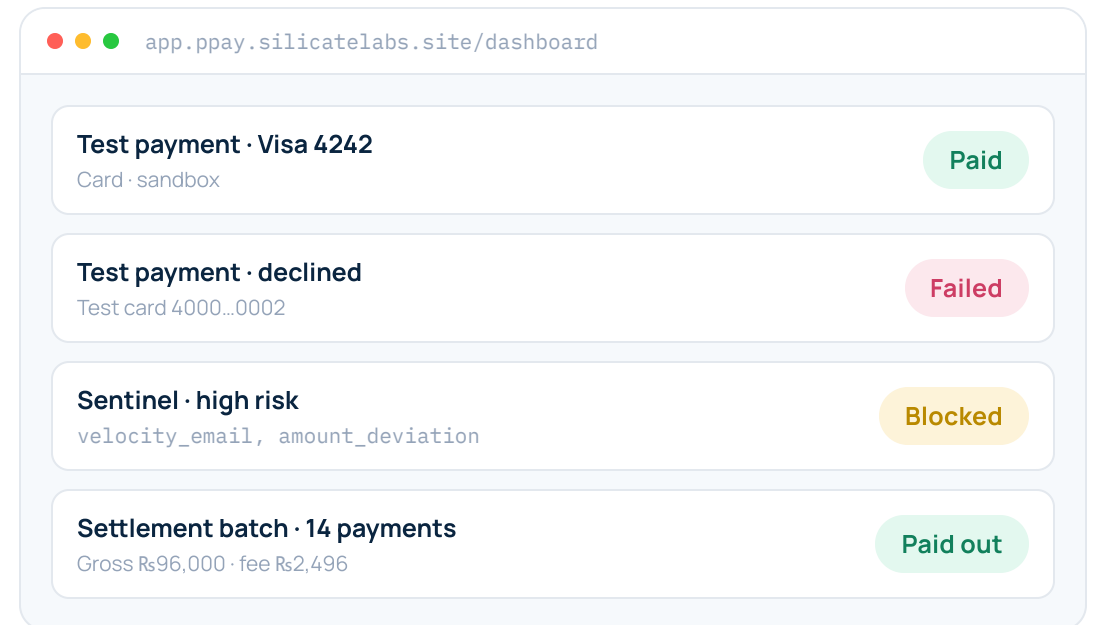
Fraud is an afterthought

Risk tooling is usually something you bolt on after a chargeback, not something that runs before the charge is attempted.

The result is that a business either accepts fewer payment methods than its customers want, or maintains four integrations and reconciles them by hand.

One integration. Every method. Risk checked before the rail.

- **One API, one dashboard.** Card, wallet, and bank transfer on a single hosted checkout, or against the REST API if you want your own flow.
- **Sentinel runs inside the payment path.** Six deterministic signals score every attempt. A score of 75 or higher blocks it before any rail is called.
- **PKR as a first class currency.** Amounts in paisas, settlement in PKR, no FX layer to reason about.
- **Built for the people who live in the dashboard.** Signed webhooks, an immutable event log, an audit trail on every action, role based access.



Twelve products, live in sandbox today

Checkout

Hosted page for card, wallet, and bank transfer.

Payment Links

A shareable URL. No code, no website.

Invoicing

Auto numbered bills with a hosted pay link.

Subscriptions

Recurring billing with a 3 attempt dunning schedule.

Sentinel

Fraud scoring before authorization runs.

Disputes

Chargeback responses on a 7 day clock.

Tax

Provincial sales tax estimates across five authorities.

Balances & Payouts

Itemised settlement batches on your schedule.

Analytics

Trends, per method success rates, comparisons.

Financial Connections

Micro deposit payout account verification.

Notifications

Seven categories, per category email control.

Team & Security

Four roles, TOTP two factor, device sessions.

Fraud scoring that runs before the payment rail, not after

Six signals combine into a 0 to 100 score. A blocked attempt never reaches authorization, so it never becomes a charge you have to refund. Every flag that fired is visible on the transaction. No black box.

velocity_email	4+ in 10 min
velocity_ip	6+ in 10 min
first_time_customer	no prior success
amount_deviation	5x trailing avg
high_amount_first_time	both together
recent_declines	3+ in 30 min

RISK SCORE

Blocked

82 / 100 · threshold 75

```
"status": "failed",  
"failure_reason": "blocked_by_fraud_rule",  
"risk_score": 82
```

We say exactly what is built, and exactly what is not

PPay runs in sandbox. It does not hold a PSO or EMI licence and does not custody funds, so no real money moves yet. Rather than hide that, we made it a product principle: every page in the app and on the marketing site labels its own status.

Live in sandbox

Twelve products, an immutable event log, signed webhooks with retries, settlement batching, and the full merchant dashboard.

Being worked on

The licensed banking partnership that will custody funds, the go live review flow, and a mobile dashboard.

Not started

Real money movement, Sentinel custom rules, device fingerprinting, split payments, and card present terminals.

A payments company that overstates what it does is a payments company you cannot trust with money.
The discipline is the pitch.

Cross mode data leakage is a structurally impossible bug

- **Schema separated by mode.** Sandbox and production data live in separate Postgres schemas, swapped per request. There is no shared table for a missed filter to forget.
- **Idempotency is required, not optional.** Checkout session creation demands an Idempotency-Key, so a retried request can never double charge.
- **No delete policy.** Records are archived, never destroyed, with an audit log entry on every account affecting action.
- **Events are immutable.** An event is written whether or not any webhook endpoint was listening, so history never depends on configuration.

STACK

API	FastAPI, async SQLAlchemy 2.0
Database	PostgreSQL 16, Alembic
Dashboard	React, Vite
Website	Static HTML, no build step
Deploy	GCP VM, nginx, systemd

Nightly backups

Rate limiting

Secrets management

Restore runbook

What actually shipped

12

Products live in sandbox, each with its own page and docs

6

Fraud signals scored on every checkout attempt

5

Provincial tax authorities modelled for sales tax on services

4

Team roles, each with its own permission boundary

3

Payment methods on one hosted checkout page

3

Dunning retries before a subscription is marked unpaid

7

Notification categories with independent email control

0

Features claimed on the site that are not actually built

Built for how Pakistan actually sells

Ecommerce and retail

Send shoppers to a checkout that already handles card, wallet, and bank transfer, instead of maintaining three integrations.

SaaS and subscriptions

Recurring billing where a failed renewal retries on a schedule rather than silently churning the customer.

Creators and freelancers

Most sales here start in a WhatsApp thread. A payment link turns that conversation into a paid order with no website at all.

Agencies and services

Bill a named client for delivered work with an auto numbered invoice and a hosted pay link.

Marketplaces and platforms

Itemised settlement batches so you can reconcile which seller each payment belonged to.

Nonprofits and fundraising

One reusable link on a poster or a QR code, taking payments from many donors.

The path from sandbox to real money

Now

Licensed partner arrangement

Structuring the partnership with a bank or PSP that will hold funds. PPay stays the software layer and never custodies money itself.

Now

Go live review flow

The KYC and account review that gates a merchant moving from sandbox to live. There is deliberately no code path that auto approves real money processing.

Next

Sentinel custom rules and device fingerprinting

A no code rule builder with per account thresholds. Today's threshold is fixed and today's signals use IP address only.

Later

Marketplaces, split payments, and in person

Routing one payment to multiple recipients, and card present collection through terminals and QR.

Built by Silicate Labs

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Product, backend, and dashboard end to end, from the payment orchestration engine to the sandbox first design principle behind every page.

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Shapes what PPay builds and in what order, working across product direction, the fraud engine, and compliance tooling.

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Try it in sandbox today.

Test cards, test wallets, and Sentinel in action. No approval needed.

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